

# **The Conical Cat-and-Mouse Engine**

*A Nested-Cone Variant of the Coaxial Rotary Engine, and Why It  
Lives or Dies on Sealing*

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## Status of This Document

This is a concept paper about a mechanism. Nothing described here has been built, tested, patented, funded, or reviewed; there is no prototype, no team, no simulation, no customer, and no commercial arrangement of any kind. Earlier drafts of this work were dressed up as a funded development programme, complete with staff, legal opinions, customer letters, and a tested demonstrator — none of which existed. Those drafts have been discarded. What remains is the only thing that was ever real: an engine geometry, an honest account of the one problem that decides whether it can work, and a description of the experiment that would settle it.

## 1. The Mechanism

Rotary internal-combustion engines come in a few families. One is the cat-and-mouse, or scissor, engine: two vanes on coaxial shafts chase each other around a shared annular chamber, rotating in the same direction but speeding up and slowing down relative to one another. As a trailing vane closes on a leading one, the gas trapped between them is compressed; it ignites, expands, and drives them apart; the cycle repeats four times per revolution without a single reciprocating part. The idea is old and much-patented (for example Galin et al., WO2016201490, and a class of earlier designs), and it has never displaced the piston, for one reason this paper will come to.

The variation proposed here is narrow and specific: replace the flat vanes sweeping a toroidal chamber with nested cones. Each rotating element carries a conical face, and the two cones close on one another along their matching taper rather than edge-to-wall. The claimed advantage is geometric. A cone settling into a cone is self-centring — radial loads push it toward alignment rather than toward one wall — and the combustion volume at the instant of ignition is compact and roughly conical rather than the long thin sliver a Wankel forms, which is thermodynamically kinder. Whether that geometric advantage is real enough to matter is one of the things a test would have to show; it is asserted here, not demonstrated.

## 2. Where It Sits in the Prior Art

Honesty about novelty matters, because the cat-and-mouse engine is a crowded, long-worked field, not open ground. It is one of the four recognised categories of rotary internal-combustion engine; Galin's coaxial-vane patent is real and current, earlier cat-and-mouse patents exist, and there is a published performance evaluation of the type. The Wankel is the only rotary to reach mass production, and it did so despite — not because of — its sealing behaviour. LiquidPiston has spent well over a decade reinventing rotary geometry and has prototypes but not volume production, which is itself a useful measure of how hard this ground is.

Against that background the honest claim is not “a genuinely novel engine with no prior art”, which an earlier draft asserted and which is false. The honest claim is narrower: the specific nested-cone interface, used for self-centring in a cat-and-mouse layout,

was not found in the patent searches behind this paper. That is a variant within a known class, and its value is entirely contingent on the problem in Section 3.

### 3. The Problem That Decides Everything: Sealing

Every rotary engine lives or dies on sealing, and this one dies on it unless something clever is done. In a piston engine, rings seal a moving piston against a fixed wall; the sealing surfaces move slowly relative to each other. Here, both cones move, and during compression they move at different speeds — the seal must hold combustion gas across a gap between two surfaces that are themselves sliding past one another at high speed, at temperatures past 1500 degrees Celsius. A 300-millimetre arm at 2000 rpm puts the interface speed in the region of sixty metres per second. Sealing an expanding-gas boundary under those conditions is the single hardest thing about the whole concept, and it is the reason the cat-and-mouse engine, for all its mechanical elegance, has never won.

Two routes exist, and this paper is honest that both are unproven. The first is a physical or gas-film seal: a sacrificial hard coating, or a deliberately maintained thin film of gas that holds the cones apart and does the sealing, in the way a gas bearing does. Whether a gas film can be both stiff enough to seal combustion pressure and stable enough not to collapse is an empirical question no amount of argument settles. The second route abandons physical sealing entirely: an “open chamber” in which the cones never quite touch and the seal is done by the gas’s own inertia and turbulence during the brief compression window — accepting some leakage as the price of removing the material-science barrier. This is more speculative still. The concept has no honest future until one of these two is shown to work; everything else is secondary.

### 4. What Is Plausible Versus What Is Unknown

It is worth separating the parts that borrow from established engineering from the parts that do not, because the earlier drafts blurred them.

| Aspect                                                    | Standing                                                               |
|-----------------------------------------------------------|------------------------------------------------------------------------|
| Igniting fuel in a rotating frame                         | Established — gas turbines burn fuel in moving/rotating flow routinely |
| Fuel delivery via rotary union to a rotating tip          | Established — rotary unions and injectors are off-the-shelf            |
| Centrifugal scavenging (flinging exhaust out by rotation) | Sound in principle — used in some turbine combustors                   |
| Self-centring from conical nesting                        | Plausible geometric claim — not demonstrated; asserted here            |
| High-speed cone-on-cone combustion sealing                | UNSOLVED — the make-or-break problem; both proposed routes unproven    |
| Open-chamber turbulence sealing                           | Speculative — no reference case; would need direct measurement         |

### 5. The Experiment That Would Settle It

The right first step is not a running engine and certainly not a company; it is a measurement of the one unknown that governs everything. A non-firing rig — a pair of representative cones driven at operating speed and heated to operating temperature by external means, with no combustion — could measure directly whether a gas film between the cones is stiff enough to hold a representative pressure without the surfaces contacting, and whether it stays stable. That is a bench experiment, not a programme. It has a clean pass/fail: if the gas film cannot hold the pressure with adequate margin, the sealed variant is dead and only the open-chamber gamble remains; if the open-chamber leakage under a simulated compression pulse is too large to sustain useful compression, the whole concept is dead and the result is worth publishing so no one else spends years on it. Only if the sealing question survives is there any point drawing a real engine, let alone building one.

Every performance number an engine like this might eventually claim — power density, vibration, efficiency — is downstream of that measurement and meaningless before it. This paper deliberately makes no such claims.

## 6. Invitation

The useful collaborator is a mechanical or tribology engineer with access to a high-speed, high-temperature test rig, willing to run the sealing measurement of Section 5 and report what it shows. A negative result is as valuable as a positive one and more likely; either way it converts a decades-old open question about this engine class into data. Nothing in this paper should be read as more than a proposal, and nothing in it has been built or arranged.

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## Appendix A: Document Metadata

|                       |                                                                                                                                                                            |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Document Title</b> | The Conical Cat-and-Mouse Engine                                                                                                                                           |
| <b>Subtitle</b>       | A Nested-Cone Variant of the Coaxial Rotary Engine, and Why It Lives or Dies on Sealing                                                                                    |
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| <b>Origin Context</b> | Honest rebuild; earlier drafts fabricated a funded development programme (staff, legal opinions, customer letters, a tested demonstrator) that did not exist — all removed |
| <b>Classification</b> | PUBLIC                                                                                                                                                                     |
| <b>Status</b>         | RELEASED                                                                                                                                                                   |

## Appendix B: Glossary

| Term                           | Definition                                                                                                          |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Cat-and-mouse (scissor) engine | A rotary engine with two coaxial vanes that alternately close and separate to form combustion chambers              |
| Gas-film / gas bearing         | A thin pressurised gas layer that separates and can seal two surfaces without solid contact                         |
| Open chamber                   | A sealing approach relying on gas inertia and turbulence rather than physical contact, accepting some leakage       |
| Rotary union                   | A coupling that passes fluid from a stationary supply into a rotating part                                          |
| Scavenging                     | Clearing exhaust gas from a chamber; here, by centrifugal action                                                    |
| Wankel                         | The only rotary internal-combustion engine to reach mass production; a triangular rotor in an epitrochoidal housing |

## Appendix C: Methodology and Transparency Statement

### Author Context and Expertise

This work is grounded in 30 years of multi-sector experience in Systems Engineering. The author is not a combustion or tribology specialist; the sealing question at the heart of this paper would need one to answer.

### Digital Methodology and Accessibility

In the spirit of transparency, I utilised a suite of Generative AI tools — specifically Claude, Google Gemini, and ChatGPT — to assist in the production of this paper. These tools were employed as “force multipliers” for data synthesis and editorial accessibility. As a writer with dyslexia, I leverage these models as assistive technology to refine grammar and streamline sentence structure.

### Verification and Integrity

Earlier versions of this work were written as an “investment ready” funded programme. They named a hired deputy investigator, legal opinions from named law firms, an independent due-diligence engagement, customer letters of intent from named companies, secured insurance, and a built and tested demonstrator. None of that existed — no person was hired, no firm was engaged, no letter was obtained, nothing was built or tested. All of it has been removed. Nothing in this paper has been built, tested, patented, funded, or reviewed, and no external person or organisation has been contacted regarding it. The prior art it cites (the cat-and-mouse engine class, the Wankel, LiquidPiston) is real; the engine variant proposed here has not been implemented.

## Appendix D: References

| Citation                           | Reference                                                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Galin et al.,<br>WO2016201490A1    | "Electromagnetic-only vane coordination of a cat-and-mouse engine." Coaxial-vane rotary IC engine patent, 2016.                                        |
| US8511276B2                        | "Cat-and-mouse type internal combustion engine, and its correlation-type crank." Prior-art rotary engine patent.                                       |
| Cat-and-mouse<br>performance study | Published performance evaluation of a cat-and-mouse type rotary engine (engine design and performance).                                                |
| LiquidPiston                       | LiquidPiston, Inc. — rotary-engine development company; prototypes without volume production. <a href="http://liquidpiston.com">liquidpiston.com</a> . |
| Wankel                             | Standard reference: the Wankel rotary engine — triangular rotor, epitrochoidal housing, apex-seal sealing challenges.                                  |